Chapter 13 exercises • I/O systems

Build state/variable tables before consulting the separate answers.

OS13-E01 • PIO publication and lost wakeups

Use the three-register, single-submitter model. The waiter tests done.

1. Give an interleaving exposing old DATA if CPU sets READY first.
2. Give a lost wakeup from checking done before sleeping. Specify the lock's scope.

OS13-E02 • Partial vector writes

Iovec lengths are[120,80,300]. First writev returns 170; a retry on the remainder returns 130.

1. Give base displacements and lengths after each result.
2. If the next result is EAGAIN, should the original vector be resent?

OS13-E03 • Virtual timers

Now4 ms, deadlines[7(A),10(B),10(C)], one one-shot hardware timer.

1. Insert6(D), then cancel A. Trace subsequent hardware deadlines.
2. Must a user thread execute each callback exactly at its deadline?

OS13-E04 • DMA cancellation and reclamation

Request8 is using DMA. Timeout requests abort, then normal completion and abort acknowledgment arrive in that order.

1. Why cannot timeout immediately unmap/free?
2. Define and prove at-most-once terminal publication.
3. May the winning completion immediately free the request control block?

OS13-E05 • Finite double-buffer batch

Four blocks take3 ms input and5 ms processing each. Two buffers forbid CPU reads before input completes and device overwrites while CPU owns a slot.

1. List input/processing intervals.
2. Compare one/two-buffer total time and the steady-state bottleneck.

OS13-E06 • STREAMS watermarks

Ordinary data queue capacity 1200 B, high 900, low 400, initial 600, messages 300 B.

Chapter 13 exercises • I/O systems (continued)

1. After one acceptance stops upstream, when does300 B service back-enable it?
2. With two upstream messages, is total buffering only300 B?

OS13-E07 • Resource bounds and Little's relation

Requests8 KiB; device60000/s; CPU10 µs/request; memory2 GiB/s and24 KiB traffic/request. Actual steady completion50000/s, response400 µs, service100 µs.

1. Calculate each resource bound and the system upper bound.
2. Find average requests in system, service, and waiting.
3. With70000 arrivals/s, no dropping, and at most60000 completions/s, can L remain20?

OS13-E08 • Naming, LBA, and durability

Blocks4096 B, sectors512 B. Read100 B at offset 12345; logical 3 maps to partition block 90; partition startLBA 1024.

1. Find displacement, whole-disk LBA, and covered range.
2. Does buffered write returning 100 prove power-loss safety?
3. If printing finished but acknowledgment was lost, does retrying a host job ID guarantee exactly-once printing?

Lin Cing https://github.com/xinhi6716/computer-science-self-study

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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